CPEN 416 / EECE 571O

Lecture 04: Projective measurements; introducing multi-qubit systems

Tuesday 16 September 2025

Announcements

- Quiz 2 today
- Some midterm information to be posted later this week

Last time

We played with *Pauli rotations*.

	Math	Matrix	Code	Special cases (θ)
RZ	$e^{-i\frac{\theta}{2}Z}$	$\begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{pmatrix}$	`qml.RZ`	$Z(\pi), S(\pi/2), T(\pi/4)$
RY	$e^{-i\frac{\theta}{2}Y}$	$\begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$	`qml.RY`	$Y(\pi)$
RX	$e^{-i\frac{\theta}{2}X}$	$\begin{pmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$	`qml.RX`	$X(\pi), SX(\pi/2)$

Last time

We explored the effect of single-qubit unitary operations on the Bloch sphere.

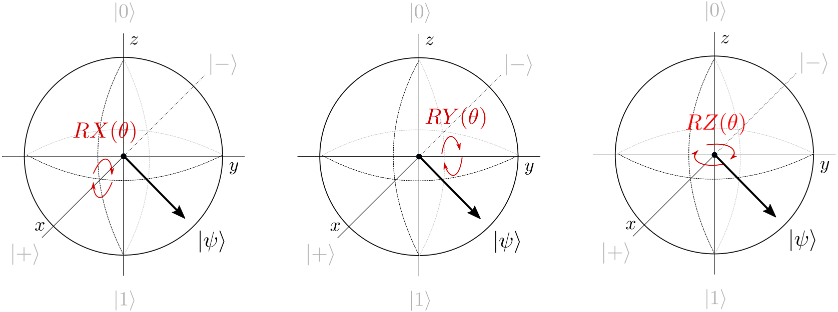


Image credit: Codebook node I.6

Last time

We introduced the “bra” part of braket notation

We defined the inner product to quantify *overlap* (similarity) between quantum states.

Learning outcomes

1. Perform a projective measurement in the computational basis
2. Apply basis rotations to perform a projective measurement in any basis
3. Mathematically describe a system of multiple qubits
4. Describe the action of common multi-qubit gates

Inner products

Exercise: compute the inner product of the state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

with itself.

Inner products

Exercise: compute the inner product between all possible combinations of $|0\rangle$ and $|1\rangle$.

$\langle 0\|0\rangle$	
$\langle 0\|1\rangle$	
$\langle 1\|0\rangle$	
$\langle 1\|1\rangle$	

Orthonormal bases

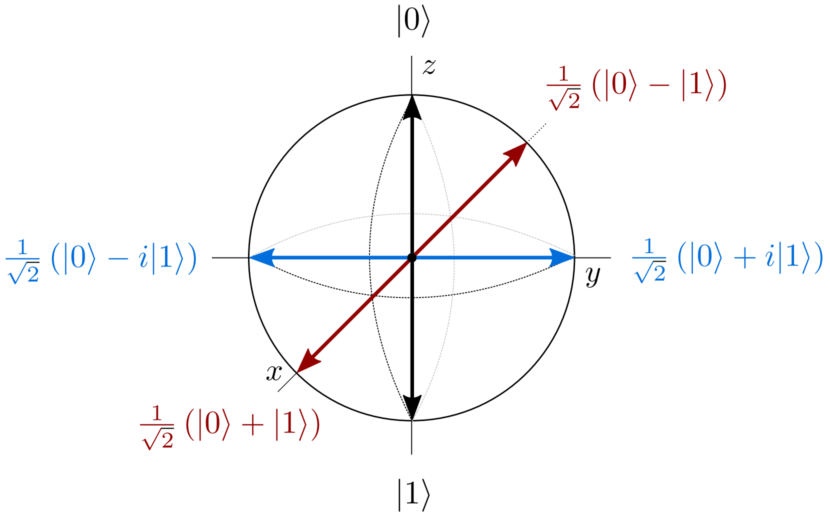
For a single qubit, a pair of states that are **normalized** and **orthogonal** constitute an **orthonormal basis** for the Hilbert space.

Exercise: do the states

$$|p\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle, \quad |m\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle$$

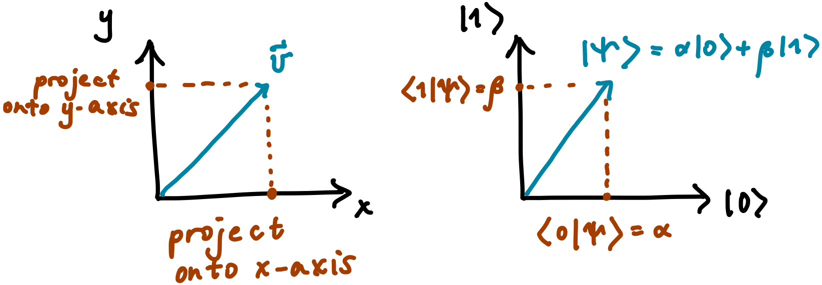
form an orthonormal basis?

Bases on the Bloch sphere



Projective measurements

Measurement is performed with respect to a basis; we perform **projections** to determine the overlap with a given basis state.



(Image for expository purposes only!)

Image credit: Xanadu Quantum Codebook I.9

Projective measurements

When we measure a qubit in state $|\varphi\rangle$ with respect to basis $\{|\psi_i\rangle\}$, the probability of observing it in state $|\psi_i\rangle$ (outcome i) is

If we observe outcome i, the qubit *remains in* $|\psi_i\rangle$.

Example: Let $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. If we measure $|\psi\rangle$ in the computational basis,

Measurement in the computational basis

Exercise: what are the measurement outcome probabilities for

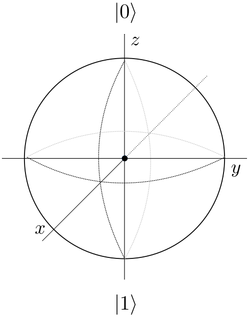
$$|p\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle, \quad |m\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle$$

in the computational basis?

Measurement in the computational basis

We know these are *not* the same state: there is a relative phase.

$$|p\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle, \quad |m\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle$$



How to tell them apart with a measurement?

Basis rotations

So far we've seen 3 types of measurements in PennyLane:

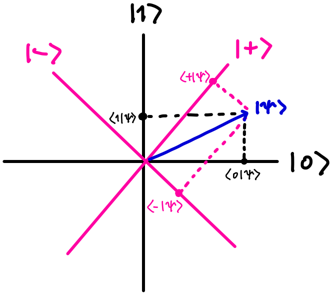
1. `qml.state()`
2. `qml.probs(wires=x)`
3. `qml.sample()`

But these all perform a computational basis measurement. This is also the only measurement allowed on most hardware.

How can we measure with respect to *different bases*?

Basis rotations

Unitary operations preserve length *and* angles between orthonormal quantum states (prove on practice A1!).



Projective measurements can be performed with respect to a different basis by applying a **basis rotation** to remap its states to computational basis states.

Image credit: Codebook node I.9

Basis rotations

Exercise: Suppose we want to measure in the "Y" basis. First, determine a sequence of gates that sends

$$\begin{aligned} |0\rangle &\rightarrow & |p\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + i|1\rangle\right) \\ |1\rangle &\rightarrow & |m\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle - i|1\rangle\right) \end{aligned}$$

Basis rotations

At the end of our circuit, apply the reverse (adjoint) of this transformation to rotate *back* to the computational basis.

If we measure and observe $|0\rangle$, we know the qubit was previously $|p\rangle$ in the Y basis (analogous result for $|m\rangle$).

Adjoints

PennyLane can compute adjoints of operations *and* entire quantum functions using `qml.adjoint`:

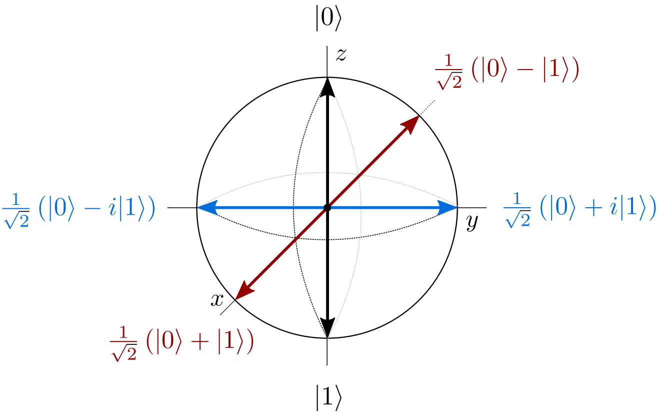
```
def some_function(phi):
    qml.RX(phi, wires=0)

def apply_adjoint(phi):
    qml.adjoint(qml.S)(wires=0)
    qml.adjoint(some_function)(phi)
```

`qml.adjoint` is a special type of function called a **transform**.

Projective measurements and the Bloch sphere

Projective measurement corresponds to a geometric projection onto the axis represented by a basis.



H-RZ exercise, revisited

Exercise: suppose we run the following circuit, but perform the measurement in the Y basis.

1. Using the Bloch sphere as intuition, what do you think the outcome probability of $|p\rangle$ looks like as a function of θ? Discuss with a neighbour.
2. Test your predictions in software.

Tensor products

Hilbert spaces compose under the *tensor product*.

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}.$$

The tensor product of A and B, $A \otimes B$ is

$$A \otimes B = \begin{pmatrix} a\begin{pmatrix} e & f \\ g & h \end{pmatrix} & b\begin{pmatrix} e & f \\ g & h \end{pmatrix} \\ c\begin{pmatrix} e & f \\ g & h \end{pmatrix} & d\begin{pmatrix} e & f \\ g & h \end{pmatrix} \end{pmatrix} = \begin{pmatrix} ae & af & be & bf \\ ag & ah & bg & bh \\ ce & cf & de & df \\ cg & ch & dg & dh \end{pmatrix}$$

Multi-qubit states

Qubit state vectors also combine under the *tensor product*:

Exercise: Determine the 2-qubit computational basis.

Question: How many n-qubit basis states? (dimension of space?)

Multi-qubit states

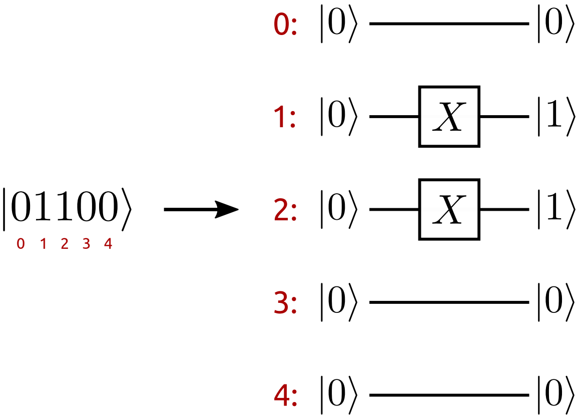
The tensor product is linear and distributive. Given

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\varphi\rangle = \gamma|0\rangle + \delta|1\rangle,$$

Question: Are tensor products of states still normalized?

Qubit ordering (very important!)

In PennyLane:



This is different in other frameworks.

Multi-qubit gates

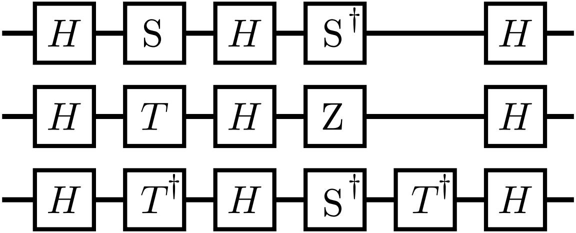
Unitary operations also compose under tensor product.

For example, apply U_1 to qubit $|\psi\rangle$ and U_2 to qubit $|\varphi\rangle$:

Exercise: determine the state of a 3-qubit system if H is applied to qubit 0, X and then S are applied to qubit 1.

Multi-qubit gates

The circuits we've seen so far only involve single-qubit gates:



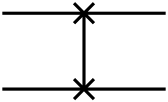
Surely this isn't all we can do...

Image credit: Xanadu Quantum Codebook I.11

SWAP

Define an operation that swaps the state of two qubits. First, specify action on basis states:

Circuit element:

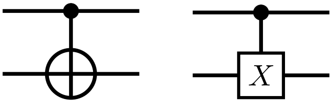


PennyLane: `qml.SWAP`

CNOT

CNOT = “controlled-NOT”. A NOT (X) is applied to second qubit only if first qubit is in state $|1\rangle$.

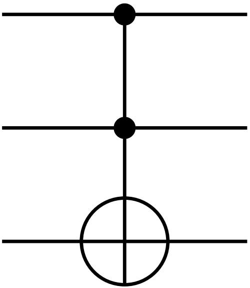
Circuit elements:



PennyLane: `qml.CNOT`

Toffoli (TOF)

There are also gates on more than two qubits, like the Toffoli gate, which is a controlled-CNOT, or controlled-controlled-NOT.

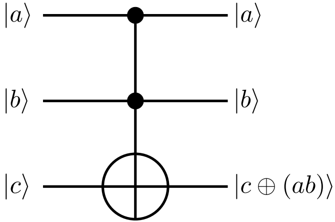


$$TOF|000\rangle =$$
$$TOF|001\rangle =$$
$$TOF|010\rangle =$$
$$TOF|011\rangle =$$
$$TOF|100\rangle =$$
$$TOF|101\rangle =$$
$$TOF|110\rangle =$$
$$TOF|111\rangle =$$

PennyLane: `qml.Toffoli`

Quantum circuits and Boolean functions

The Toffoli implements a reversible AND gate. It is also universal for classical reversible computing.

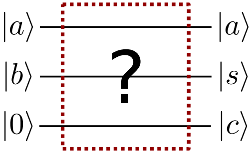


Exercise: what Boolean function does CNOT implement?

Quantum circuits and Boolean functions

X, CNOT, TOF can be used to create Boolean arithmetic circuits.

Exercise: use a combination of these gates to create a quantum version of a one-bit adder. Test it in PennyLane on all possible bit values of *a* and *b*.



Next time

Content:

- Entanglement
- Measurement of multi-qubit states
- Bell basis measurements

Action items:

1. Work through last section of yesterday's tutorial (bases)
2. Practice assignment 1 (can do Qs 1, 2, parts of 4)

Recommended reading:

- For today: Codebook IQC, SQ, MQ; N & C 1.2, 1.3.1-1.3.5, 2.2.3-2.2.5, 4.3
- For next time: Codebook SQ MQ (all tied up); N & C 1.3.6, 1.3.7, 2.3